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Mobile phone component object detection algorithm based on improved SSD

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Abstract

Aiming at the problem of low accuracy of small object detection and poor system robustness in the detection of mobile phone component image targets, an improved SSD algorithm is proposed to detect mobile phone component image targets. This article proposes an algorithm, CSP-SSD(Cross Stage Partial SSD). By improving the network structure, using gradient flow information, using deconvolution and using multi-scale transformation, the traditional SSD algorithm is improved, so that it can better adapt to the detection task of mobile phone components. The test results show that when the image size is 300*300, the mAP of the algorithm reaches 78.9%. The proposed method has the characteristics of high detection accuracy for small objects.

Keywords: Target detection; CSPNet (Cross Stage Partial Network); deconvolution; SSD;

1. Introduction

In industrial production lines, many products have numerous parts and complex inspection processes. Due to the objective factors, the method of manual contact detection on the production line is prone to human error, and the efficiency is low, it is easy to cause product damage. These errors can lead to the failure of component and a significant impact on subsequent production, and also resulting in unpredictable losses. In this case, it is necessary to adjust the inspection process in the industrial production line, reduce labor costs, improve efficiency, and avoid device damage caused by collisions during manual inspection.

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The current common target detection methods for mobile phone components are mainly divided into manual detection methods and deep learning target detection methods. Traditional detection methods use artificial extraction, and then calculate the relevant information of the image and the target based on the extracted features. This type of method is susceptible to environmental and artificial interference, and the detection effect of mobile phone component targets is poor. Target detection based on deep learning mainly uses large servers, massive data, and detection algorithms to detect mobile phone components through neural networks[15]. This method is not affected by man-made environments and has high detection accuracy for mobile phone components.

As we all know, deep learning is a transformative technology in machine learning, especially in computer vision. With the advantages of Convolutional Neural Networks[16] in extracting high-level features of images, deep learning has achieved great success in the field of target detection. In 2014, Girshick et al. first proposed the deep learning model R-CNN[17] based on convolutional neural network on CVPR2014. In 2015, Girshick and Ren et al. respectively proposed FastR-CNN[1] and Faster R-CNN[2] algorithms, and their mAP on the PASCAL VOC2012 dataset was increased to 68%. In 2016, Redmon et al. proposed the YOLO[3] algorithm. The detection speed is 10 times faster than Faster R-CNN[4]. The extracted candidate regions are directly classified and regressed in the main network structure, and it will classify and locate The integrated thoughts provide new ideas for subsequent research[18]. Based on the YOLO algorithm, in 2016, LIU and Redmon proposed the SSD[6] algorithm. The SSD algorithm simplifies the entire process of object detection, integrates target determination and recognition, and greatly improves the operating speed[19]. In 2020, YOLOv4 and YOLOv5 have been released one after another, introducing new data enhancement methods, and the detection speed and accuracy are greatly enhanced.

In this paper, an improved SSD algorithm is used to detect mobile phone components, and feature extraction of mobile phone component images through deep neural networks. Aiming at the solving problems of low target recognition accuracy, difficulty in training, large memory usage, the reuse of gradient information is used to alleviate the need for a large number of inference calculations and improve the network learning ability. For the poor detection of small targets, deconvolution is used to fuse semantics. The experimental results show that compared with the classic single-stage detection algorithm, the algorithm in this paper has a higher detection accuracy.

2. Related Work

2.1. SSD Algorithm Structure

SSD is a single-stage target detection algorithm, using VGG16[5] as the backbone network, which is a multi-scale feature map target detection algorithm. The SSD algorithm uses feature maps of different scales to detect targets of different sizes, and select candidate frames of different scales for different feature maps in the subsequent pyramid[20].

2.2. DSSD Algorithm Structure

DSSD[7]adjusts the backbone network, using ResNet101 as the feature extraction network, and adds some Residual modules to enhance the feature expression of the original model, The purpose of deconvolution is to fuse deep feature maps and shallow feature maps, so that the fused feature maps have both the semantic information of the deep feature maps and the shallow features location information of the map.

2.3. Spatial-Channel Parallelism Algorithm Structure[11]

The network can be regarded as an upgraded version of DenseNet, which improves the information flow of dense blocks and transition layers, optimizes the gradient back propagation path, improves the learning ability of the network, and improves the processing speed and memory. In terms of target detection, lightweight design has also been made.

2.4. Analysis of The Disadvantages of The SSD

Although the SSD algorithm maintains a high level of accuracy and speed, it still has some shortcomings: SSD Backbone uses VGG16, which is not enough for deep-level information learning and expression. Although the feature map contains more semantic information, the edge information is insufficient, and it is not friendly enough for small target detection. In the SSD model, the feature map that really detects small targets is only the bottom Conv4_3 layer, and because this layer is in the shallow layer of the convolutional network, the nonlinearity of feature is not enough, which leads to a certain degree of feature loss and easy leakage. Therefore, the detection effect of the SSD model on medium and small targets is weaker than that of large targets. This paper analyzes the shortcomings of the SSD model for small and medium target detection, and proposes an improved CSP-SSD model based on split gradient flow[14].

3. The Concrete Realization of The Improved CSP-SSD Model

In this paper, the CSP-SSD model is improved based on the SSD model. The backbone network is the VGG16 network. In view of the repeated use of gradient information in the SSD model, the calculation bottleneck is too large and requires more cycles to complete the inference problem. This article uses the split gradient flow to integrate the gradient changes from beginning to end into the feature map to enhance the learning ability of CNN. Effectively improve the utilization of each computing unit, thereby reducing unnecessary energy consumption.In order to overcome the shortcomings of low-level feature layers, this paper adds Conv3_3 as the bottom feature map to the feature pyramid to detect smaller-scale targets. In view of the problems that the SSD model has layer-by-layer convolution and pooling, which cause some features to be lost, this paper designs a new feature fusion structure to perform deconvolution feature fusion to make up for the shortcomings of insufficient semantic information and insufficient edge information of feature maps.

3.1. Split Gradient Glow

The goal of training a neural network is to optimize the cost function, the cost function need to find a global minimum or a local minimum[8]. No matter what gradient descent algorithm is used, the gradient of each parameter needs to be calculated first. In VGG, the output of the i-th layer is the input of the i+1th layer. The formula can be expressed as:

$$\begin{aligned} x_1 &= w_1 * x_0 + b_1 \\ x_2 &= w_2 * [x_0, x_1] + b_1 \\ x_k &= w_k * [x_0, x_1, ..., x_{k-1}] + b_k \end{aligned} \tag{1}$$

If the backpropagation algorithm is used to update the weights, the weight update formula can be expressed as:

$$\begin{aligned} w_1' &= f(w_1, g_0) \\ w_2' &= f(w_1, g_0, g_1) \end{aligned} \tag{2}$$

f is the weight update function, g_i representing the first convolutional layer in the i-th gradient propagate. It showed that, in update the weights of different convolutional layers, most of the gradient information is repeatedly used, therefor the same information of gradient will be used repeatedly in different convolutional layers[12] .

The architecture of CSPResNet[14] is as follows:

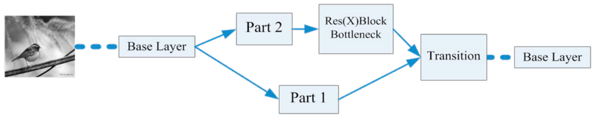


Fig. 1. The Architecture of CSPResNet

In each stage, according to the channel: $x_0 = [x_0^{'}, x_0^{''}]$,the feature map of the base layer of is split into two parts .The first part $x_0^{'}$ is directly connected to the end of the base layer, and the second part $x_0^{''}$ passes through the ResNet block. The steps of partial transition layer are as follows: the transition layer will be passed through in the output of ResNet layers ;then, the results which transition layer output will be used in another layer and passed through another transition layer. The feedforward channel and weight update formulas of CPSResNetNet[12] are as follows:

$$\begin{aligned} x_k &= w_k * [x_0, x_1, \ldots\ldots, x_{k-1}] + b_k \\ x_T &= w_T * [x_0, x_1, \ldots\ldots, x_k] + b_T \\ x_Y &= w_Y * [x_0, x_T] + b_Y \end{aligned} \tag{3}$$

$$\begin{aligned} w_k^{'} &= f(w_k, g_0^{''}, g_1, g_2, \ldots., g_{k-1}) \\ w_T^{'} &= f(w_T, g_0^{''}, g_1, g_2, \ldots., g_k) \\ w_Y^{'} &= f(w_Y, g_0^{'}, g_T) \end{aligned} \tag{4}$$

It can be found that the gradients from base layers are integrated separately. In addition, the feature maps that have not passed the ResNet block are also independently integrated. As for the gradient information used for weight update, the two parts do not contain the redundant gradient information of the other party.

3.2. Multi-Deconvolution

In order to solve the lack of semantic information in the low-level feature maps in the SSD model and solve the lack of detailed information in the high-level feature maps, the improvement method of the DSSD[9] model is to perform deconvolution of the high-level feature layers and merge the low-level feature layers. This paper proposes the idea of multi-deconvolution, which integrates the deconvolution into three steps, and adds a new feature layer Conv3_3 to extract edge information.

Multi-Deconvolution increases the resolution of the feature map and expands the receptive field[13]. Before feature fusion, the feature layer with small scale needs to be deconvolved to unify the resolution of the feature map, and the same resolution as the feature map to be fused. Deconvolution operation calculation formula: d=s(i-1)+k. In the formula, s represents the step size, k represents the size of the convolution kernel, d is the size of the deconvolution output feature map, and i is the size of the input feature map. Before performing feature fusion in this paper, according to the deconvolution feature map scale calculation formula, the designed deconvolution operation parameters are shown in Table 1:

Table 1. Network parameterss

Input	Input Scale	Output Scale	Output
Conv11_2	1 * 1	3 * 3	Dcov11
Co_conv10_2	3 * 3	5 * 5	Dcov10
Conv9_2	5 * 5	7 * 7	Dcov9_1
Dcov9_1	7 * 7	8 * 8	Dcov9_2
Dcov9_2	8 * 8	10 * 10	Dcov9
Co_conv8_2	10 * 10	20 * 20	Dcov8_1

3.3. Model Architecture

The CSP_SSD model structure is shown in Figure 2 (where ○represents the feature fusion module).This model starts deconvolution from the Conv11_2, Conv9_2, and Conv7 feature maps and merges with the two lower-level feature maps, so that the feature map has the high-level of feature map in the semantic information and the detailed information of the low-level feature map. The specific convolution structure is shown in the figure below. Firstly, Conv11_2 is deconvolved and Conv10_2 is fused to generate the detection feature layer Co_conv10_2, and Co_conv10_2 is deconvolved and fused with Conv9_2 to generate the detection feature layer Co_conv9_2. Conv9_2 is deconvolved and fused with Conv8_2 to generate the detection feature layer Co_conv8_2, then Co_conv8_2 is deconvolved and fused with Conv7 Generate the detection feature layer Co_conv7. Conv7 is generated by Conv6_2. The CSPResNet module is introduced to divide the gradient flow information. Conv6_2 is divided into two parts, one part is directly connected to the end of the base Layer, the second part passed through the 3 layer ResNet block, finally, output to Conv7 through a transition layer, and Conv7 is reversed. The product is fused with Conv4_3 to generate the detection feature layer Co_conv4_3, and then Co_conv4_3 is deconvolved and fused with Conv3_3 to generate the detection feature layer Co_conv3_3 to complete the deconvolution operation.

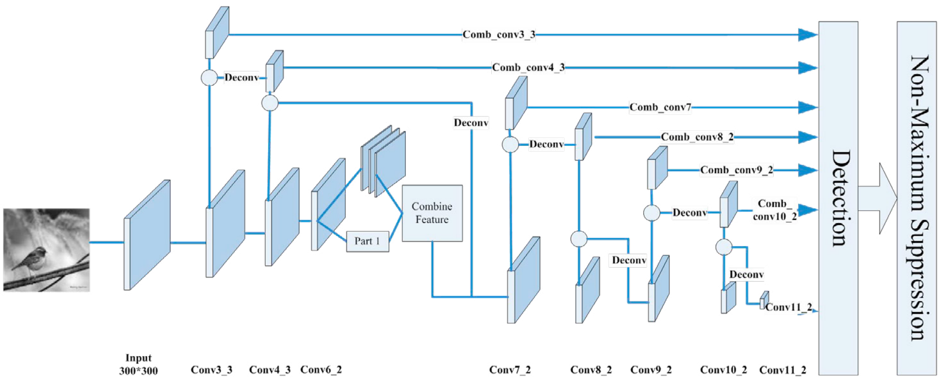


Fig. 2. CSP_SSD model structure diagram

4. Comparative Experiment And Result Analysis

4.1. Experimental Data

The dataset used in the experiment is the dataset collected on the actual production line. The training dataset has a total of 533 images divided into 5 categories. The number of cameras is greater than 600, the number of flashes is greater than 610, the number of cards is greater than 1066, the number of screws is greater than 1599, and the number of batteryBay is greater than 523. In order to solve the problem of a small number of samples, rotated the image angle to expand the number of samples, adjusted hue, changed saturation, etc.

4.2. Training and Experimental Results Analysis

The experiment uses ubuntu18.04 system, Caffe deep learning framework, the hardware configuration is Intel (R) Core TM i9-9900K CPU@3.60GHz processor, 32G running memory, NVIDIA GeForce GTX 2080Ti graphics card.

The network parameters are shown in Table 2:

Table 2.Deconvolution Parameters

Hyperparameter	Value
Learning rate	0.001
Gamma	0.5
Step	6000,8000,11000,12000
Batch	32
Mometum	0.9
Max_iter	13000

On the public data set VOC2007 test, mAP reached 79.0%, which was an increase of 1.7% and 0.4% compared to the original model.the target detection accuracy are shown in Table 3.

Table 3.Comparison of the new model with the original model

Method	Model	mAP
SSD300	VGG16	77.3%
DSSD321	ResNet	78.6%
New model	VGG16	**79.0%**

Under the model in this paper, the final result realizes that the mAP on the actually collected dataset PhoneVOC reaches 80.8%, which is an increase of 8.4% compared to the original SSD model, and 0.4% compared to the original DSSD model. The comparison results of the network model in different categories of target detection accuracy are shown in Table 3.

Table 4.Comparison results of object detection accuracy in different categories of different algorithms

Method	Model	mAP	camera	flash	batteryBay	screws	card
SSD300	VGG16	72.4%	88.7%	69.3%	89.0%	25.3%	89.6%
DSSD321	ResNet	66.1%	86.5%	69.3%	**94.1%**	14.2%	**92.1%**
New model	VGG16	**78.9%**	**90.0%**	**81.2%**	89.2%	**49.2%**	84.7%

From the comparison results of detection accuracy, it is not difficult to see that the New model proposed in this article has improved the detection of camera, flash, and screws, especially the small target screws. The improvement is obvious. Due to the small number of categories, the value of a single category will directly affect the final result. For small target screws, because the pixels are small, the use of continuous deconvolution will cause the loss of feature information and the addition of noise information, so after fusion, The characteristic of the feature map information is interfered by noise information, making the detection effect of the DSSD model not obviously, which is even lower than that of SSD. Therefore, Multi-deconvolution is used to avoid the problem of excessive noise information in the deconvolution process of the DSSD model. The final experiment proved that fusing the gradient information into the feature map helps to improve the learning ability of the model, and proved the semantic information of the high-level feature map and the edge information of the low-level feature map. Insufficient information extraction also avoids the problem of excessive noise information. Calculated the maximum proportion of each category. As shown in Table 5, batteryBay and card are large targets, camera and flash are medium targets, and screws are small targets.

Table 5.Percentage of pixels in different object

Input	batteryBay	card	camera	flash	screws
Percentage	<=80%	<=13.0%	<=7.5%	<=3.5%	<=0.8%

As shown in Figure 3, this experiment selected five target categories: batteryBay, camera, card, flash, and screws for comparison. The square represents the detection accuracy of the CSP-SSD algorithm in this article for the five target categories. Compared with the SSD300 algorithm represented by the circle and the DSSD321 algorithm represented by the diamond, the accuracy has been significantly improved, which proves that the addition of the lower-level fusion feature layer and the adjustment of the scale have a significant effect.

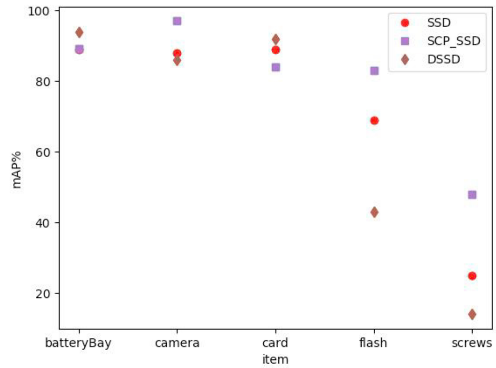
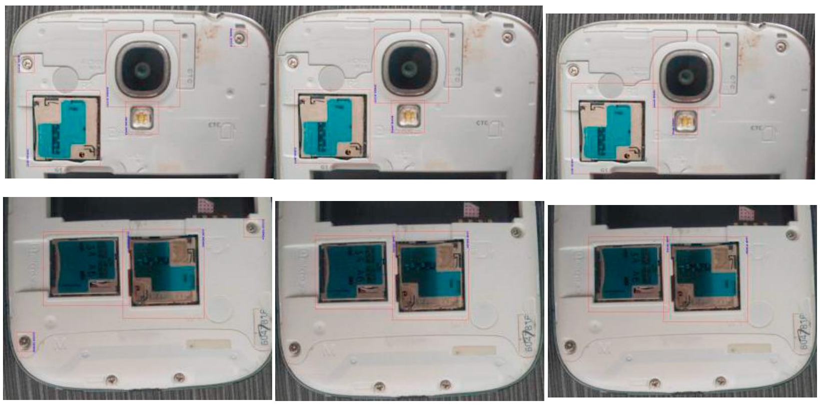


Fig. 3. Comparison chart of object classes-mAP.

4.3. Comparison of Detection Results

Figure 4 is a comparison diagram of some pictures in the SSD300 algorithm model, as well as in the DSSD321 algorithm model and the CSP-SSD algorithm model. From the detection results of (a), (b), (c) in Figure 4, it showed that the CSP-SSD target detection algorithm proposed in this paper is improved based on the SSD300 algorithm and DSSD321 algorithm, which is more friendly to small target detection than the SSD300 target detection algorithm. Small targets detection is greatly improved, and it is not only more friendly to small target detection in complex scenes, but also the higher recognition accuracy of large and medium targets, and the target positioning is also more accurate.



(a) CSP-SSD (b) SSD300 (c) DSSD321

Fig. 4. Comparison of test results.

5. Concluding Remarks

In order to further improve the detection effect of the target, this paper proposes the CSP-SSD algorithm with gradient flow information and deconvolution to improve the SSD algorithm. The backbone network uses VGG16, and the network structure is redesigned to increase the semantic information and edge information of the feature map. Without adding too much noise, the idea of gradient flow is introduced, and lower-level fusion feature maps are added for repeated detection of small targets. Finally, the model achieved a score of 78.9% in mAP on the test set. The detection of small targets is improved significantly. On the basis of the algorithm model of this article, the next step can be an in-depth study on how to carry out a lightweight network and realize the operation of terminal equipment.

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